9. (i) Find

$$\int \frac{(3x+2)^2}{4\sqrt{x}}\,\mathrm{d}x \qquad x > 0$$

giving your answer in simplest form.

(5)

(ii) A curve C has equation $y = \mathrm{f}(x)$.

Given

- $\mathrm{f}'(x) = x^2 + ax + b$ where a and b are constants
- the y intercept of C is -8
- the point $P(3, -2)$ lies on C
- the gradient of C at P is 2

find, in simplest form, $\mathrm{f}(x)$.

(6)